

DMV302 Data Mining and Visualizations

‘Data Visualization’



Component code and name	DMV302 Data Mining and Visualization
Assignment name	Simulation of data mining and visualization techniques and Presentation. Source Codes + 5-10-minute presentation
Weighting	40%
Submission deadline	Week 16
Week issued	Week 12

Brief

This assessment requires you to complete three data mining tasks:

- **Task 1:** Create visualization diagrams on the house sales dataset programmatically. You need to be creative and programmatically visualize a data set and are required to deliver a 5–10-minute presentation explaining your choices of the graphs, discussing the patterns, insights, and information you have drawn from your visualization, and reflecting on your experience in this task.
- **Task 2:** Detecting anomalies in the weather data.
- **Task 3:** Association analysis on transactional data.

Please refer to the Task Instructions (below) for details on how to complete this task.

Task Instructions

Task 1: Data Visualization and Clustering (30%)

Import the data file titled `DMVA3DT1.csv`. Each column of this file contains 81 explanatory variables describing (almost) every aspect of a residential home.

- (a) Implement a Python program that reads in the `DMVA3DT1.csv` file and programmatically creates interactive data visualization plots by making suitable choices of selecting the correct style of graphs/plots for the sales distribution. Make a careful selection of some features from the dataset that majorly contribute towards sales price changes, such as *house_style*, *building_type*, *year_of_sold*, etc.
- (b) Implement a Python clustering program to create clusters using 3D scatter plots
- Use any efficient clustering algorithm (K-means, variants of K-means, DBSCAN, Agglomerative).
 - Set the number of clusters to any optimized value (Hyper-tuned).
 - Draw a 3D scatter diagram illustrating the three clusters. Use different shapes and colors to distinguish data points in each cluster. Label your diagram appropriately.

Save the source code as `'DMVA3_Task1.ipynb'`.

Task 2: Anomaly detection and data visualization (35 %)

Import a weather data (.csv) file titled `DMVA3DT2.csv`. This file contains the city's weather history data for a whole financial year (1/7/2022 to 30/6/2023). The table below sets out the column numbers and briefly describes each column.

Column number	Description
A	The date of the record; formatted dd/mm/yy
B	The actual average temperature for that day
C	The actual minimum temperature for that day
D	The actual maximum temperature for that day
E	The average minimum temperature on that day
F	The average maximum temperature on that day
G	The lowest temperature on that day
H	The highest temperature on that day
I	The actual amount of rain on that day
J	The average amount of rain on that day
K	The highest amount of rain on that day

(a) Detect data anomalies in the dataset using the following techniques:

- LOF (Local Outlier Factor)
- Isolation Forest
- One-Class SVM.

Display the anomalous outcomes from each technique as labels or scores.

(b) **Be as creative as possible.** Create an interactive diagram(s) to visualize the given data set and anomalies found using the above different approaches to capture as much **information** and **insights** as possible in the form of graphs using Python libraries (**not just dumping data on a graph**).

- Label your diagram or diagram(s) appropriately.
- There is no requirement as to the number of diagrams you should create. However, you should aim to use as few diagrams as possible, each capturing as much information as possible.

Save the source code as 'DMVA3_Task2.ipynb'.

(c) Prepare a 10-12-minute presentation to demonstrate your findings, and all the diagrams you created in **Task 1a, 1b, and Task 2b**. In your presentation, you should:

- Explain your diagram or diagrams and why you chose to visualize the data set in the way you did. Your discussion should be connected to the characteristics of the data set.
- Discuss the insights, knowledge, patterns, or information that is obtained directly from your visualizations.
- Reflect on your experience in this task and discuss the data challenges.

Task 3—Association analysis (20 %)

Download the dataset file titled `DMVA3DT3_Shoppingtransactions.csv`. This file contains transactions, each row representing a member ID, date, and items purchased in one transaction. Here are some sample rows:

	Member_number	Date	itemDescription
0	1808	21-07-2015	tropical fruit
1	2552	05-01-2015	whole milk
2	2300	19-09-2015	pip fruit
3	1187	12-12-2015	other vegetables
4	3037	01-02-2015	whole milk

Implement the following steps to perform association analysis:

1. Group items by "Member_number" and "Date" and create a list of items purchased by each member on a specific date.
2. Drop unimportant columns and convert the DataFrame to a list.
3. Defining threshold values for parameters `Support`, `Confidence`, and `Lift`.
4. Train the Apriori, FP-growth, and ECLAT models on the transformed dataset.
5. Design a simple command-based interface using three models that allow users to interact with the application. Users will be prompted to the menu with the following options until the user selects Option 6.
 1. Option to select 1-2 items.
 2. Select those items from a drop-down menu.
 3. Choose the Algorithm (Apriori, FP-Growth, and ECLAT).
 4. Display the results of that item as Left-side->right-side, `Sup[item]`, `Con[item]` and `lift[item]`.
 5. exit

Option 1: Allows for finding the association between one or two items.

Option 2: Allows the addition of items. Depending upon the item selection in Option 1, you can add one item or two items.

Option 3: Choose the Algorithm type.

Option 4: Calculate and print the support of an item or a set of two items. For example,

```
Burger-> {Spaghetti,} Sup=0.17
```

OR

```
{Burger, Spaghetti} -> {parmesan cheese, sausage,..} Sup=0.16
```

Calculate and print the confidence of an item or two items, provided that another item or two items were bought. For example:

```
Cookies--> {Chocolate,} con=0.18 OR (sausage, whole milk)->(yogurt,fruit)  
con=0.20
```

Calculate and print the lift of an item or two items, provided that another item or two items are bought. For example:

```
Cookies--> {Chocolate,} lift=0.18 OR (sausage, whole milk)->(yogurt,fruit)  
lift=0.20
```

Save your Python code in a file called 'DMVA3_Task3.ipynb'.

Assessment instructions

You will need to submit the source files in a single zip folder with appropriate names. Here are the instructions:

- You may use any software/device to prepare your presentation. However, it is highly recommended that you use PowerPoint or equivalent software.
- You are encouraged to give a presentation either in person or must show your face during the online presentation for authentication purposes.
- Your presentation must be within 5-10 minutes with clear volume with no background noise.

All Python programs in this assessment should comply with the following requirements:

- Follow Python naming conventions. If in doubt, refer to the *Google Python Style Guide* at <https://google.github.io/styleguide/pyguide.html>;
- Be implemented with Python 3 (**NOT** Python 2. x);
- Be implemented with any IDE of your choice;
- Contain adequate and useful comments; and
- Be thoroughly tested and do not contain any syntax or logical errors.

Referencing

Students must use the APA style for citing and referencing research appropriately. Please see more information on referencing at <https://mediadesignschool.libguides.com/apa>.

Submission Guidelines

Naming conventions

DMV302LastNameFirstNameBrief3.zip



- Task 1 (Folder)
 - DMVA3_Task1.ipynb
- Task 2 (Folder)
 - DMVA3_Task2.ipynb
 - Presentation .ppt/.pptx

- Task 3 (Folder)
 - DMVA3_Task3.ipynb

Submission structure

Submit the code file via the Assessment 3 link in the main navigation menu in DMV302: Assignment 3, by the end of Week 16.

Assessment Criteria

The brief outline of assessment criteria is as follows. For more details on each section, check MyLearn assessment rubrics.

1. **Data Visualization and Clustering (30%)**
2. **Anomaly detection and Data visualization (35%)**
3. **Association Analysis (20%)**
4. **Effective presentation (5%)**
5. **Programming Practices (10%)**

Submission Policy

Submit this task via the **Assessment 3** link in the main navigation menu in DMV302 Data Mining and Visualization. Please refer to the Submission Policy on MyLearn for further details.

You are required to submit **Python code files (.ipynb)** and **a presentation (.ppt)**. Please ensure that you submit individual files that are zipped/compressed into one folder. Before submitting your assignment, you should check it against the detailed rubric criteria on Mylearn to ensure that you have satisfactorily addressed all the criteria that will be used to mark your assignment.

Academic Integrity

I declare that, except where I have referenced, the work I am submitting for this assessment task is my work.

I have read and am aware of Media Design School's Academic Integrity Policy and Procedure, viewable online at <https://www.mediadesignschool.com/policies-and-procedures>.